

A. S. M. G. B. AGALAWATTE

Hardware & Embedded Systems Engineer · PCB Design

Kandy, Sri Lanka | gamunuagalawatte@gmail.com | +94 71 059 9507

gamunuagalawatte.github.io/portfolio | github.com/gamunuagalawatte | linkedin.com/in/gamunu-agalawatte

PROFESSIONAL SUMMARY

Electrical & Electronic Engineering graduate specializing in embedded systems, PCB design, and hardware–software integration. Experienced in taking products from prototype to field-deployed, mass-production-ready hardware, with hands-on work in ESP32/RP2040 platforms, custom multi-layer PCB design, firmware debugging in C, and Python desktop tooling. Competition-winning robotics background. Seeking a role in embedded or hardware engineering where I can build systems that physically exist.

TECHNICAL SKILLS

PCB Design	EasyEDA Pro, KiCad (developing), multi-layer (2–4 layer) design, schematic capture, layout & routing, bare-IC boards, power regulation, DFM
Power & RF	Qi wireless charging receiver design (BQ51013B, BQ25100), coil selection & power budgeting, nRF52840 BLE systems, RF antenna integration
Embedded	RP2040, ESP32, ESP8266, SIM800L, ICM-20602 IMU, USB CDC, SPI, I ² C, UART, sensor interfacing
Firmware	C / C++ (Arduino, PlatformIO), fuzzy-logic control, firmware debugging & validation
Software	Python, PySide6 / Qt6, multithreading, event-driven & layered architecture, JSON protocol design, Git
Modelling & Tools	MATLAB / Simulink, LabVIEW, AutoCAD, Fusion 360, SquareLine Studio, VS Code
EE Core	Power electronics, digital systems, control systems, signal processing, instrumentation

EXPERIENCE

Electronics Engineering Intern

Jun 2025 – Sep 2025

Sri Lanka Telecom PLC — Digital Labs, Head Office, Colombo 01

- Took an early-stage prototype through full product development to a field-deployed, mass-production-ready device, owning PCB design, firmware validation, and on-site testing.
- Designed a production-grade PCB around an ESP32 microcontroller and SIM800L GSM module for a remote field-monitoring system.
- Debugged and validated embedded firmware in C, resolving communication and hardware-interface issues ahead of deployment.
- Diagnosed customer-reported faults on deployed devices, isolating hardware from configuration issues and identifying incorrect field tuning as the root cause rather than hardware failure.

Tech: ESP32 · SIM800L · Custom PCB · Embedded C · System validation

Engineering Trainee

Jun 2024 – Sep 2024

DIMO (Pvt) Ltd — Power Solutions Department, Colombo 08

- Supported annual site inspections and maintenance of medium-voltage switchgear, step-down transformers, and Ring Main Units (RMUs).
- Assisted with insulation resistance and overcurrent relay testing using primary and secondary injection kits on MV panels and transformers.
- Observed Factory Acceptance Tests (FATs) on Siemens 33 kV cast-resin transformers and NXAIR H switchgear; prepared technical inspection reports and utility bid documentation.

Tech: MV switchgear · Transformers · Insulation & relay testing · FAT procedures

PROJECTS

JumperZ Studio — Smart Programmable Breadboard

Sep 2025 – Jun 2026

Final Year Project · 3-person team · SLIIT — Desktop Software & Physical Design Lead

A programmable breadboard platform built around a custom RP2040/ESP32 mainboard — a 12× CH446Q analog switching matrix driving 400 individually addressable WS2812B LEDs across a 6-layer PCB, with dual INA219 sensors for live current and voltage sensing — reconfigured entirely from a desktop application, removing manual jumper wiring.

- Sole developer of the desktop application (~150 Python modules): designed a six-layer, domain-driven architecture (Core, Domain, Infrastructure, Services, UI, App) with enforced one-way dependencies.
- Built a publish–subscribe event bus decoupling hardware I/O from the UI across 6 named background threads, with a `run_on_ui()` trampoline to safely marshal all event publishes onto the Qt main thread.

- Designed a USB CDC four-port communication protocol with a custom JSON command schema for real-time circuit synchronization.
 - Implemented a circuit netlist builder (Union-Find with path compression) and a Manhattan-style routing algorithm to visualize connections across a 400-node addressable LED matrix.
 - Led physical system design: dual-board PCB outlines in AutoCAD, a magnetic pogo-pin inter-board connector, and a 3D-printed enclosure.
- Tech:** Python · PySide6 (Qt6) · Multithreading · USB CDC · JSON protocol · arduino-cli · AutoCAD · Git
- GitHub:** github.com/gamunuagalawatte/jumperz-studio

Line Following Robot — Custom PCB

2025

Autonomous robot on a self-designed 4-layer PCB

- Designed a 4-layer PCB (dedicated VCC, GND, and two signal layers) using bare ICs, with RP2040 as the main MCU.
 - Integrated ESP8266EX Wi-Fi (with antenna), ICM-20602 IMU, USB Type-C, dedicated power regulation, and a CJ2301 P-channel MOSFET for automatic battery/USB-C power switching.
 - Built a 9× TCRT5000 IR sensor array and motor-control subsystem; implemented fuzzy-logic line-tracking firmware.
- Tech:** EasyEDA · RP2040 · ESP8266EX · ICM-20602 · Fuzzy logic · 4-layer PCB
- GitHub:** github.com/gamunuagalawatte/RP2040-line-following-robot

Maze-Solving Micromouse (V1 & V2)

2023 – 2025

Team project · Mechanical structure & sensor design lead

- An autonomous maze-solving robot using Flood Fill path planning and PID motor control on a dual-core ESP32-S3 — Core 1 handles path planning, Core 0 handles real-time motor control — with PID tuning modelled in MATLAB. Built across two iterations, each improving speed, mechanical precision, and turn accuracy.
- V1: Led the full mechanical structure and sensor placement for a competition-grade micromouse with dual-MCU architecture; designed a 2-layer mainboard.
 - V2: designed a compact 4-wheel geared drivetrain — 1200 RPM motors at a 3:2 gear ratio, achieving a top speed of 0.924 m/s — and sensor geometry for accurate turn detection.
 - **Achievements:** Champion — MicroMaze 2025 (IIT); 1st Runner-Up — Robofest 2025 (SLIIT), 30 s fast run; finalist / 8th place — Robofest (V1).
- Tech:** PlatformIO · Arduino · EasyEDA · Fusion 360 · Dual-MCU
- GitHub:** github.com/gamunuagalawatte/Micromouse

Autonomous Poultry Feeding System

2023

Group project · ESP32-based automated feeder with touchscreen UI

- Developed an ESP32-based system to feed poultry automatically by schedule, quantity, and pit position, with a multi-screen touchscreen interface built in SquareLine Studio.
 - Created sensor calibration and position-tracking routines and designed the 3D-printed mechanical parts.
- Tech:** ESP32 · TFT UI · SquareLine Studio · 3D printing

EDUCATION

BSc (Hons) Electrical & Electronic Engineering

2022 – 2026

Sri Lanka Institute of Information Technology (SLIIT) · Second Upper Class (expected)

GCE Advanced Level

Trinity College, Kandy · Combined Mathematics, Physics, Chemistry — B, B, C

GCE Ordinary Level

Trinity College, Kandy · 9 Distinctions (A)

ADDITIONAL INFORMATION

Languages

English, Sinhala (Native)

REFERENCES

Prof. Eranga Wijesinghe
 Assistant Professor, Department of Electrical and Electronic Engineering
 SLIIT | **Office:** +94 11 754 3950 | **Mobile:** +94 74 153 1013
sites.google.com/view/aiai-lab-usj/home

Ms. Thanushika Jathunga
 Senior Lecturer, Department of Electrical and Electronics Engineering
 SLIIT | **Mobile:** +94 71 371 6733 | thanushika.j@slit.lk